Message Dissemination in Intermittently Connected D2D Communication Networks

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***Abstract*—Device-to-device (D2D) communications enable direct communications and information distribution among closely located devices in wireless networks. Many applications of D2D communications require message dissemination to a group of mobile users at certain locations. The challenges of message dissemination come from highly dynamic network environments due to movements of devices. Existing studies on message dissemination have focused on information propagation speed and latency, but the *size of the area* affected by message dissemination at time t is also critical to D2D communication applications that heavily depend on message dissemination among users in a *geographic region*. In this paper, we study the fundamental issues in D2D communications: how far the message dissemination can reach by time t (referred as *dissemination distance*) and how long the dissemination takes to inform nodes located at distance d (referred to as *hitting time*), especially in dynamic, intermittently connected networks. We first derive analytic bounds of dissemination distance and hitting time under different dissemination mechanisms, providing the spatial and temporal limits of message dissemination. Analytic results are further validated by simulation results of several corresponding dissemination algorithms. Finally, two application scenarios are provided to illustrate how our results serve as guidelines to choose or design appropriate dissemination methods for different D2D communication applications.**

***Index Terms*—Device-to-device (D2D) communication network, message dissemination, dissemination distance, hitting time.**

I. Introduction

DEVICE-TO-DEVICE (D2D) communication is commonly used for direct communication between two closely located devices in wireless networks. D2D communication underlaying a cellular infrastructure [2], [3] has been proposed to take advantage of the physical proximity of communicating devices, offloading the cellular system, increasing bit-rate, and improving cellular coverage and robustness to infrastructure failures. Moreover, D2D communication is the foundation of mobile *ad hoc* network (MANET), which enables users of wireless devices to create instant, *ad hoc* networks to interact with people geographically nearby to them.

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Applications and services based on D2D communication can be manifold [2]. As an example, people at events, such as a football match or concert, can anonymously chat and share promotional materials, photos, or videos with each other. When two or more phones are within transmission range of each other, they can form an *ad hoc* network without connection to a Wi-Fi or cellular network. D2D communications can provide an efficient solution for downloading and distributing information among mobile phones. As another example, vehicular *ad hoc* network (VANET) has emerged to improve road safety, traffic efficiency, and driving convenience through vehicle-to-vehicle (V2V) communications. Emergency information, such as traffic collision warning, can be disseminated to drivers by multi-hop V2V communications. In applications of D2D communications, message dissemination is an essential function for information distribution and sharing among mobile devices, where devices are mostly carried by mobile human or installed on vehicles driven by human.

Consequently, device mobility (i.e., human mobility) introduces major challenges for message dissemination in D2D communication networks, including rapid network topology changes, frequent link and path breaks, and intermittent network connectivity. These challenges hinder the deployment of D2D communication applications that are heavily dependent on message dissemination among geographically nearby mobile devices. Therefore, one fundamental issue of achieving D2D communication applications is to understand the performance of message dissemination.

Many researchers have studied the performance of message dissemination including propagation speed and latency [4]–[11]. But how large is the area affected by message dissemination is also critical. Many applications based on D2D communication target a group of mobile users in specific geographic regions, such as audience in a stadium and drivers on a road of an accident. Nonetheless, spatial propagation receives the least attention [12] and is mainly performed in 1-Dimensional (1-D) VANETs with static or simple node mobility (e.g., constant speed or moving direction) [13]–[16]. Therefore, the fundamental yet open issues to be addressed are: how far a dissemination can reach by time t (referred as *dissemination distance*) and how long a dissemination takes to inform partial or all nodes within an area of radius d (referred as *hitting time*). Especially, we study these issues in D2D communication networks with dynamic network topology and frequently fragmented network connectivity due to node mobility. Answers to these questions can provide understanding of the spatial limit of dissemination methods and the time needed to inform users at certain distance from the source. The results can also yield insights in whether and how the application requirements (such as allowable latency) could be possibly satisfied, thus help network designers choose appropriate dissemination strategy according to application requirements, especially in highly dynamic and intermittently connected D2D networks.

We start with investigating device mobility and dissemination algorithms because of their crucial impact on dissemination performance. A mobile device can only move to adjacent points around its previous location at each step due to both geographic constraints and speed limit of its carrier (i.e., human or vehicle). Hence, we consider a *constrained* mobility model that node v_i's location at time $t+1$ is distributed in a region centered at its location at time t with radius a. Denote relays that are actively rebroadcasting and spreading the message as *disseminators*. Apparently, only disseminators contribute to message propagation while other recipients carrying latent messages do not contribute to spreading the message. Thus, dissemination performance depends on how many disseminators are used and how they are chosen. We define a general dissemination algorithm—*L-copy*$(L \geq 1)$ message dissemination—that maintains L disseminators actively spreading the message. A succeeding disseminator can be chosen from neighbors of the preceding disseminator *direction-invariantly* or based on *geographic information* in order to spread out the message quickly. Based on whether geographic information is used to assist dissemination, we classify L-copy message dissemination into direction-invariant and geographic-assisted dissemination strategies. We will study dissemination distance and hitting time under both types of dissemination strategies.

In a dissemination process, disseminators solely determine the performance of message dissemination (i.e., spatial and temporal limits of dissemination). Message dissemination happens where the disseminators retransmit the message as they move around and change from one mobile device to another. By shrouding the changes of disseminators from one mobile device to another, we focus on the locations of the messages carried by disseminators—referred as *active messages*, i.e., where the message transmissions are. Therefore, we formulate the mobility of *active message* that includes movements of active message carriers (i.e., disseminators) and jumps incurred by transmissions from proceeding disseminators to succeeding disseminators. Without specifying relay nodes on information propagation path, message mobility can focus on where active messages reach rather than by which nodes carry and relay.

Based on the formulation of *active message mobility*, we then derive lower and upper bounds for the farthest distance that active messages reach at time t and the first hitting time that active messages reach devices at distance d from the original source location under L-copy direction-invariant and geographic-assisted dissemination strategies, respectively. Simulation results show that several well known dissemination algorithms, including stateless opportunistic forwarding (SOF) [17] and GPS-based broadcasting (GBB) [18], are well bounded by our analytic bounds. Both analysis and simulation evidence that regardless of the number of disseminators used in the dissemination, the upper bound on expected dissemination distance increases with the *square root* of time t in direction-invariant dissemination strategy, while it increases approximately *linearly* with time t in geographic-assisted dissemination strategy. In addition, two realistic scenarios in VANETs—*post-crash warning* and *emergency vehicle signal preemption*—along with analytical and simulation results, suggest that for D2D communication applications that target an area near the source location, dissemination algorithms with multiple disseminators are suitable, while for applications that target an area far from the source location, geographic-assisted dissemination strategy is preferable in order to satisfy application requirements.

II. Related Work

Many researchers have studied the performance of message dissemination in various application scenarios of D2D communications, such as MANETs and VANETs. Essentially, most researchers are interested in (1) how *fast* the information can be spread and (2) how *far* the dissemination can propagate. Hence, the metrics of message dissemination performance can be in time domain (e.g., dissemination latency and propagation speed), and space domain (e.g., propagation distance) [12].

In the time domain, message dissemination latency and information propagation speed have been studied in MANETs [4]–[7], [19], [20] as well as in VANETs [8]–[11], [13], [16]. Both Zheng [4] and Xu [5] derived limits of information diffusion rate in large-scale wireless networks, which showed a constant upper bound on propagation speed. [7] presented a graph based model to characterize connectivity properties and derived a scaling law for message delay in large scale disconnected *ad hoc* networks. Kong and Yeh [6] found that delay scales linearly with the Euclidean distance between the sender and the receiver when the network is in the subcritical phase, and the delay scales sub-linearly with the distance if the network is in the supercritical phase. Paper [20] analyzed the information propagation speed limits in large scale mobile and intermittently connected networks (i.e., Delay Tolerant Networks (DTNs)). Jacquet *et al.* [19] derived the asymptotic capacity and delay in large scale mobile networks.

Because real D2D communication network has finite size and number of nodes in the network, nonasymptotic results on performance of message dissemination are also analyzed. Specially, the latency and speed of message dissemination in VANETs—a very important application of D2D networks—have received lots of attention. Fracchia and Meo [10] analyzed the average delay, the probability that a vehicle is informed, and the average number of duplicate messages received by a vehicle within a area of interest in a 1-D highway scenario. Similarly, paper [8] developed upper and lower bounds for the time of information propagation between two nodes in a 1-D network and showed that more vehicles on the road does not necessarily promote the fast propagation of information. The relationship between latency and reliability is studied in [11]. Jacquet *et al.* [9] analyzed information propagation in bidirectional vehicular DTNs. The authors proved and computed a threshold of vehicle density, above which information speed increases dramatically over vehicle speed, and below which information propagation speed is on average equal to vehicle speed.

On the contrary, the space metric domain receives less attention [13]–[16]. Paper [13] derived spatial propagation of information in a 1-D vehicle-to-vehicle ad-hoc network. Both analysis and simulation show that information propagation depends on vehicle traffic characteristics, such as vehicle density, average vehicle speed, and relative speed among vehicles. Resta *et al.* [14] derived lower bounds on the probability that a car at distance d from the source of the emergency message correctly receives the message within time t. In a 1-D static network scenario, the authors show that besides d and t, this probability depends also on 1-hop channel reliability and the message dissemination strategy. Further considering node mobility, [15] analyzed the propagation distance in 1-D VANETs with constrained vehicle mobility. Jacquet *et al.* [16] showed that when the vehicle density is smaller than a threshold, routing using bidirectional traffic in bidirectional vehicular DTNs provides a gain in the propagation distance, which follows a sublinear power law as time elapses.

In addition, research on spatial propagation is mainly performed in 1-D VANETs with static or simple node mobility (e.g., constant speed or moving direction). Nonetheless, spatial propagation is an important performance metric as many D2D applications target information delivery in geographic regions (i.e., geocast), such as collision warning and regional weather forecast. There still lacks understanding of the spatial propagation properties in 2-D networks with realistic node mobility, which we will study in this paper.

III. Models and Problem Formulation

In this section, we introduce our network and mobility models and two general dissemination strategies, and formally define dissemination distance and hitting time.

A. Network and Mobility Models

Assume that at time 0, n nodes $\{\mathcal{X}(0)\} = \{X_1(0), \ldots, X_n(0)\}$ are uniformly distributed at random in a two-dimensional torus $\mathcal{B} = [0, B]^2$ where $B = \sqrt{n/\lambda}$, for some $\lambda > 0$. The random vector $X_i(0)$ denotes the location of node v_i at time 0. By definition [21], $\{\mathcal{X}(0)\}$ is a homogeneous Poisson point process. n nodes are Poisson distributed in the network with density $\lambda = n/B^2$ everywhere. Two nodes v_i and v_j can communicate with each other at time t if only if their distance is less than or equal to transmission range r, i.e., $d_{i,j}(t) \triangleq \|X_i(t) - X_j(t)\| \leq r$. The average number of neighbors per node is therefore smaller than or equal to $\pi r^2(n/\mathcal{B})$. Simulations in [22] suggested that six to eight neighbors can make a network connected with high probability when n is small; while Xue and Kumar [23] showed that if the average number of neighbors is smaller than $0.074 \log n$, then the network is almost surely disconnected when n is large. Therefore, we assume that λ and r are small enough that $\pi r^2(n/\mathcal{B})$ is smaller than six and $0.074 \log n$ for small and large size networks, respectively. This will capture the intermittent connectivity in D2D communication networks due to node mobility and limited radio coverage, thus enable us to study the performance of message dissemination in such networks.

Suppose that time is slotted and nodes move according to a given mobility model $M(t), t = 1, 2 \ldots$. In other words, the displacement of node v_i from its position $X_i(t-1)$ to $X_i(t)$ is distributed according to $M(t)$. We consider a generic mobility model [24] defined as follows.

Definition 1: (Generic Mobility) Given initial nodes' positions $\mathcal{X}(0)$ at $t = 0$, the spatial distribution $X_i(t)$ of node v_i at time t is around a point x_i^* by a non-increasing and direction-invariant function $\Psi_i(x) = \Psi(x - x_i^*)$. Assume that Ψ_i is non-zero only in a region characterized by a constant a; that is, $\Psi_i(x) = \Psi(x - x_i^*) > 0$ when $\|x - x_i^*\| \leq a$ and $\Psi_i(x) = \Psi(x - x_i^*) = 0$ otherwise.

This mobility model is very general as it covers a wide range of mobility processes. The case of static nodes uniformly deployed over the area can be obtained by setting $\Psi_i(x) = \delta(x - X_i(0))$. The i.i.d. mobility model in [25] corresponds to the case when $\Psi(x)$ is a constant function independent of x and $a = \infty$. When $a < \infty$ and $x_i^* = X_i(0)$, we obtain the mobility model used in [6].

To mimic device mobility that is constrained by the speed limit of device carrier and dependent on previous movements, we consider a constrained mobility model: setting $a < \infty$ and $x_i^* = X_i(t-1)$ at time slot t, $X_i(t)$ is uniformly distributed at random in $\mathcal{A}(X_i(t-1), a)$-the circular region centered at $X_i(t-1)$ with radius $a > 0$. The positions $X_i(t)$ are mutually independent among all nodes and only dependent on previous locations $X_i(t-1)$. This mobility process can also be interpreted as the following. At each time slot, a node chooses a random direction uniformly from $[0, 2\pi]$ and travels for a random length which is chosen from $[0, a]$ following certain distribution. Denote by $A(t)$ and $\theta(t)$ the length and angle of a movement step at time t, respectively. Movement vector of a node v_i at time t is $Y_M(t) \triangleq A(t)e^{j\theta(t)}$ (j is imaginary unit) with origin at $X_i(t)$ and endpoint uniformly distributed in $\mathcal{A}(X_i(t), a)$. Since both sequences $\{A(t)\}$ and $\{\theta(t)\}$ are i.i.d. and independent from each other, $Y_M(t) (t = 1, 2 \ldots)$ are also i.i.d. random variables.

Remarks 1: We use the *constrained mobility* model because (i) it generally accounts for a wide range of realistic mobility processes in D2D communication scenarios, including Manhattan mobility [26] and random walk, (ii) it reflects the speed limit of nodes (usually wireless devices carried by humans or installed on vehicles) that nodes can jump to adjacent locations with pre-assigned probabilities and each movement step is limited in a circular region around previous location, and (iii) n nodes are Poisson distributed in the network $\mathcal{B}$ with density λ everywhere at all times (Proof: see Appendix).

B. Dissemination Strategy

Dissemination performance, such as information propagation speed and dissemination latency, depends on how many nodes are recruited to disseminate the message (i.e., the number of disseminators) and how the disseminators are chosen. Existing dissemination algorithms either rely on unicast, such as geographic routing (e.g., GeRaF [27]), or broadcast (e.g., geocast routing [28], [29]). Clearly, by using as many disseminators as possible, full epidemic broadcast achieves fast dissemination, but leads to network congestion. Hence, limited-copy dissemination (i.e., limited number of disseminators) is more feasible in order to save network resources and enable the coexistence of multiple applications. If being used for choosing disseminators, geographic information has the potential to enhance information propagation speed. However, geographic information may be unavailable for devices in the network and exchanging geographic information consumes the limited network resources. Since the number of disseminators and whether geographic information is used in disseminator selection affect performance of dissemination strategies, we classify dissemination strategies according to these factors, based on which we study dissemination performance.

Definition 2: (L-Copy Message Dissemination) Assume that node v_0 initiates a message dissemination at time 0. First, the message will be spread to $L (L \geq 1)$ distinct disseminators. Then, each disseminator will independently rebroadcast the message to its neighbors until it finds its succeeding disseminator. Disseminator is selected based on criteria imposed by applications. This process repeats until the dissemination completes.

L-copy message dissemination with small L is particularly useful in the following situations: 1) network has limited capacity; 2) network load is heavy; 3) nodes are computationally-constrained or energy-constrained devices. Thus, the network could only support a small number of disseminators in order to share the limited network resources by multiple applications. On the other hand, more disseminators actively rebroadcasting the message at the same time can likely increase the speed of message spreading and enhance the probability of successful deliveries to destinations (i.e., dissemination reliability). Thus, L-copy message dissemination with large L is favorable for time critical or emergency message dissemination.

Definition 3: For a message dissemination strategy, if a disseminator chooses its next-hop disseminator isotropically (equally in all directions), we refer it as *direction-invariant dissemination*; if geographic information is used to assist disseminator selection such as to enhance dissemination performance (such as propagation speed), we refer it as *geographic-assisted dissemination*.

Direction-invariant dissemination can be implemented by imposing the receivers to probabilistically decide whether to become a disseminator; geographic-assisted dissemination can be achieved by scheduling receivers to broadcast their decision based on their locations. In either strategy, the feedback mechanism is used between two disseminators such that the previous disseminator stops broadcasting the message and the new disseminator starts spreading the message. Note that the traffic induced by feedback mechanism would not be overwhelming in intermittently connected networks.

C. Problem Formulation

In this paper, we are interested in how far a dissemination reaches at time t and how long a dissemination takes to spread the message to certain location, i.e. the spatial and temporal limits. In order to derive the spatial and temporal limits of

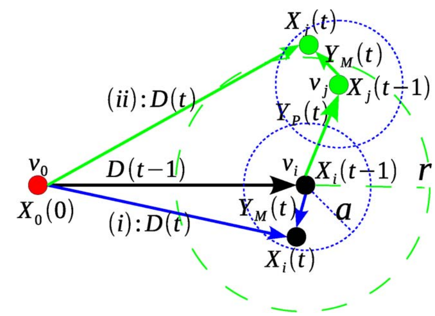

Fig. 1. Dissemination distance varies due to movements of disseminators and active message transmission from disseminator v_i (the black node) to next disseminator v_j (the green node).

message dissemination, we omit the effects of buffering or congestion, and assume that a message can be transmitted instantaneously between two nodes in range (i.e., omitting message transmission time). Under these assumptions, we are able to derive spatial and temporal bounds since they correspond to an ideal scenario with that respect. Actually, previous assumptions have little impact on the accuracy of our results because information transmission occurs much faster than the speed of the mobile nodes and message transmission time is much smaller than the dissemination latency incurred by dynamic topology and intermittent connectivity in D2D communication networks.

1) Dissemination Distance: In a dissemination, a message copy held by a disseminator is called *active* message. On the other hand, recipients carrying *latent* (i.e., non-active) message do not retransmit the message, thus do not contribute to increasing delivery ratio and reducing dissemination latency. Hence, we ignore latent messages, and focus on active messages and disseminators.

Denote by $\mathcal{V}(t)$ the set of disseminators at time t and $|\mathcal{V}(t)| \leq L$. Let us place a Cartesian coordinate system in the network with its origin at the source location. The *dissemination vector*$D^L(t)$ is the vector from source point $X_0(0)$ to the location of the farthest disseminator at time t. The length of dissemination vector is called *Dissemination Distance*, which is defined as

$$\left|D^L(t)\right| \triangleq \max_{v_k \in \mathcal{V}(t)} \left\{\|X_k(t) - X_0(0)\|\right\}. \tag{1}$$

$|D^L(t)|$ by definition is the distance from the source location to the farthest location reached by disseminators at time t. In other words, $|D^L(t)|$ is the size of the area that is affected by the dissemination at time t. For simplicity, we denote $|D(t)|$ as the dissemination distance when $L = 1$.

To avoid specifying the relay nodes, we study $D^L(t)$ through the mobility of active messages. Denote by $Y(t) = D(t) - D(t-1)$ the progress of an active message from time $t-1$ to $t (t = 1, 2, \ldots.)$. As shown in Fig. 1, suppose v_i (the black node) is the disseminator at time $t-1$, (i) if v_i is also the disseminator at time t, the active message moves with v_i, which means that $Y(t)$ equals the movement step $Y_M(t)$ of v_i; (ii) if node $v_j (i \neq j)$ (the green node) is selected as the next disseminator at time t, the active message jumps from v_i to v_j and moves with v_j, which means that $Y(t)$ equals the propagation vector $Y_P(t)$ plus the movement step $Y_M(t)$ of v_j. In L-copy message dissemination, we number each active message from 1 to L and denote by $|D_k(t)|$ the farthest distance reached by the kth active message at time t. We assume that active messages are independent of each other. Hence, the progress of the kth active message, $Y_k(t)$, equals either $Y_M(t)$ or $Y_M(t)+Y_P(t)$.

Remark 2: For *direction-invariant* dissemination, $E(Y_P^x(t)) = E(Y_P^y(t)) = 0$; for *geographic-assisted* dissemination that utilizes geographic information to increase dissemination distance $|D(t)|$, $E(Y_P^x(t)) \geq 0$ when $D_x(t) \geq 0$ and $E(Y_P^x(t)) \leq 0$ when $D_x(t) < 0$ (the same for the y-component).

2) Hitting Time: In L-copy message dissemination, to find the first time that the dissemination reaches certain location, we define *hitting time* as

$$\tau^L(d) \triangleq \inf_{t>0} \left\{t : \left|D^L(t)\right| \geq d\right\} \tag{2}$$

where d is a positive constant. Essentially, $\tau^L(d)$ is the message's fist hitting time of the region that is outside of the circular region centered at $X_0(0)$ with radius d. Denote by $\tau(d)$ when $L=1$ for simplicity.

Dissemination distance and hitting time manifest the spatial and temporal limits of message dissemination, respectively. Dissemination distance reveals the size of the zone affected by a dissemination in which nodes are at least partially informed by the message at time t. Hitting time uncovers the minimum latency of reaching nodes at d distance from the source. Putting together, they can be used to determine whether a dissemination has reached mobile devices in an area of interest and whether D2D communication can possibly satisfy the time requirements of applications. Intuitively, different dissemination strategies exhibit different performance in dissemination distance and hitting time. We expect our results to provide guidelines on choosing appropriate dissemination methods according to application requirements.

IV. Lower Bounds

To begin with, we derive lower bounds on dissemination distance and hitting time by examining a 1-copy message dissemination that the source will be the only disseminator. Originally, the source initiates a message dissemination with $|D(0)| = 0$, and actively spreads the message while all other recipients carry latent messages without retransmission. The dissemination is solely determined by the mobility of the source node. In other words, the progress of the active message from time $k-1$ to $k (k = 1, 2, \ldots)$ equals the movement vector $Y_M(k)$. Hence, at time t, $D(t) = \sum_{k=1}^{t} Y_M(k)$.

A. Lower Bound on Dissemination Distance

To find lower bound on $|D(t)|$, we first examine mobility vector $Y_M(t)$.

Lemma 1: Under constrained i.i.d. mobility, movement vector $Y_M(t)$ satisfies that $E(|Y_M(t)|) = (2a/3)$ and $E\{|Y_M(t)|^2\} = (a^2/2)$, where $|Y_M(t)|$ is the length of $Y_M(t)$ and a is the maximum movement length per time slot.

Proof: Suppose node i locates at $X_i(t-1)$ and $X_i(t)$ at time $t-1$ and t, respectively. The movement vector $Y_M(t)$ has its origin at $X_i(t-1)$ and endpoint at $X_i(t)$ that is uniformly distributed in $\mathcal{A}(X_i(t-1), a)$. The length of a movement step $|Y_M(t)| = A_i(t) = \|X_i(t) - X_i(t-1)\|$.

Let the circular region $\mathcal{A}(X_i(t-1), a)$ be partitioned to m concentric rings indexed by $j, 0 \leq j \leq m-1$, each of which is with equal ring width ϵ. When ϵ is much smaller than a,

$$P\left(A_i(t) = x\right) \approx \frac{2\pi x\epsilon}{\pi a^2} = \frac{2}{a^2} x\epsilon.$$

Hence,

$$E\left(|Y_M(t)|\right) = E\left(A_i(t)\right) = \int_0^a \frac{2}{a^2} x^2 dx = \frac{2a}{3}$$

$$E\left\{|Y_M(t)|^2\right\} = \int_0^a \frac{2}{a^2} x^3 dx = \frac{a^2}{2}.$$

■

Based on Lemma 1, we have

$$E\left(|D(t)|^2\right) = \sum_{k=1}^{t} E\left(|Y_M(k)|^2\right) = \frac{a^2 t}{2}. \tag{3}$$

Since better designed dissemination algorithm and multiple disseminators can spread out the message faster, $a^2t/2$ can serve as the lower bound of the mean square displacement (MSD) of $|D(t)|$.

B. Lower Bound on Hitting Time

We have the following lower bound on probability distribution of hitting time $\tau(d)$.

Theorem 1: Hitting time $\tau(d)$ satisfies that

$$P\left(\tau(d) < t\right) \geq \max\left\{0, 1 - \frac{4(d+a)^2}{a^2 t}\right\}. \tag{4}$$

Proof: Denote the x-component and y-component of dissemination distance vector $D(t)$ as $D_x(t)$ and $D_y(t)$ respectively. Then

$$\tau(d) \leq \tau_x(d) \triangleq \inf_{t>0} \left\{t : |D_x(t)| \geq d\right\}.$$

And $D_x(t) = \sum_{k=1}^{t} Y_M^x(k)$, where $Y_M^x(k)$ is the x-component of mobility vector $Y_M(k)$ as shown in Fig. 2.

Based on constrained mobility model, for $-a < x < a$ and small Δx, we have

$$P\left(x \leq Y_M^x(k) \leq x + \Delta x\right) \approx \frac{2\sqrt{a^2 - x^2}\Delta x}{\pi a^2} \tag{5}$$

i.e., the green area in Fig. 2. The probability distribution of $Y_M^x(k)$ is a even function, thus $E(Y_M^x(k)) = 0$. Further,

$$E\left(\left(Y_M^x(k)\right)^2\right) = 2\int_0^a x^2 \frac{2\sqrt{a^2 - x^2}}{\pi a^2} dx = \frac{a^2}{4}. \tag{6}$$

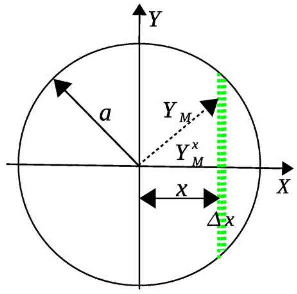

Fig. 2. Probability distribution of x-component of mobility vector Y_M.

As $D_x(t)$ is a generalized 1-D random walk with independent and mean-zero increments $Y_M^x(k)$, $D_x(t)$ is a martingale process with respect to $(Y_M^x(k), k \geq 0)$ according to the definition of martingale. According to Wald's Second Inequality, the stopping time of martingale $D_x(t)$ satisfies

$$E\left(\tau_x(d)\right) = \frac{E\left(D_x^2\left(\tau_x(d)\right)\right)}{a^2/4} \leq \frac{4(d+a)^2}{a^2}. \quad (7)$$

Based on Markov inequality $P(\tau_x(d)<t) \geq 1-(E(\tau_x(d))/t)$. Therefore,

$$P\left(\tau(d) < t\right) \geq P\left(\tau_x(d) < t\right) \geq \max\left\{0, 1 - \frac{4(d+a)^2}{a^2 t}\right\}.$$

■

V. Upper Bounds

In this section, we move to derive upper bounds on dissemination distance and hitting time under dissemination strategies that use or do not use geographic information in choosing disseminators, respectively.

A. Upper Bounds on Dissemination Distance

Originally, $D(0) = 0$; at time t, $D(t) = \sum_{k=1}^{t} Y(k)$, where $Y(k)$ is the progress of an active message from time $k-1$ to $k (k = 1, 2, \ldots)$. As shown in Fig. 1, $Y(k)$ equals either movement vector $Y_M(k)$ or $Y_M(k)$ plus propagation vector $Y_P(t)$. In order to derive dissemination distance $D(t)$, we study random variables $Y_M(k)$ and $Y_P(k)$ in Lemmas 1 and 2, respectively.

Lemma 2: Propagation vector $Y_P(t)$ satisfies that

$$E\left(|Y_P(t)|\right) \leq r\left(1 - e^{-\lambda\pi r^2}\right) \quad (8)$$

$$E\left\{|Y_P(t)|^2\right\} \leq r^2 - \frac{1}{\lambda\pi}\left(1 - e^{-\lambda\pi r^2}\right) \quad (9)$$

where $|Y_P(t)|$ is the length of propagation vector $Y_P(t)$.

Proof: Assume node v_i locates at $X_i(t)$ at time t and $N_i(t)$ is the set of v_i's neighbors. Denote by

$$S(t) = \max_{v_j \in N_i(t)} \left\{\|X_i(t) - X_j(t)\|\right\} \quad (10)$$

the maximum distance between node v_i and its neighbors at time t. Apparently, the length of propagation vector, denote by $|Y_P(t)|$, is equal to or less than $S(t)$ and the equality holds if and only if v_i chooses its farthest neighbor as next-hop disseminator. Hence, for $0 \leq x \leq r$

$$P\left(|Y_P(t)| \geq x\right) \leq P\left(S(t) \geq x\right) \quad (11)$$

i.e., random variable $S(t)$ stochastically dominates $|Y_P(t)|$.

Let $A_m(x, y)$ denote the event that there exist m neighbors in area $\{s : x \leq \|s - X_i(t)\| \leq y\}$, $(0 \leq x < y \leq r)$. As n nodes are Poisson distributed in the network with density λ everywhere at all times, $P(A_m(x, y)) = e^{-\lambda\pi(y^2-x^2)}([\lambda\pi(y^2 - x^2)]^m/m!)$. For $0 \leq x \leq r$,

$$P\left(S(t) \geq x\right) = 1 - P\left(A_0(x, r)\right) = 1 - e^{-\lambda\pi(r^2-x^2)}. \quad (12)$$

Hence

$$E\left(S(t)\right) = r - e^{-\lambda\pi r^2} \int_0^r e^{\lambda\pi x^2} dx \leq r(1 - e^{-\lambda\pi r^2})$$

$$E\left(S^2(t)\right) = \int_0^r x^2 \left(e^{-\lambda\pi(r^2-x^2)}\right)' dx = r^2 - \frac{1 - e^{-\lambda\pi r^2}}{\lambda\pi}. \quad (13)$$

Combining (11) and (13) completes our proof. ■

Remark 3: Lemma 2 shows that the largest expected length of propagation vector increases as transmission range r and node density λ increase, which provide higher probability of finding a next-hop disseminator that locates far from its previous disseminator, thus can spread out the message as fast as possible.

Because of intermittent network connectivity, an active message travels a journey in the network area through movements and transmissions of disseminators rather than propagating on a path. Hence, dissemination distance $|D(t)|$ is affected not only by distributions of $Y_M(k)$ and $Y_P(k) (1 \leq k \leq t)$ but also by the number of jumps of an active message within time t, which is defined as $\mathcal{N}(t) = \sum_{k=1}^{t} 1_{Y(k)=Y_M(k)+Y_P(k)}$. As transmission between two disseminators occurs when a preceding disseminator meets its succeeding disseminator, $\mathcal{N}(t)$ is determined by intermittent connectivity as well as dissemination algorithms. We derive an upper bound for $\mathcal{N}(t)$ by investigating the *first passage time*, which is defined as the time when two nodes first meet.

Lemma 3: In a large scale network, suppose two independent nodes v_i and v_j move according to constrained i.i.d. mobility model. Then, there exists constant $C > 0$ that the first passage time T_F of nodes v_i and v_j satisfies

$$P(T_F > t) \geq Ct^{-\alpha}, \text{ for all sufficiently large } t \quad (14)$$

where α is a constant determined by mobility model and $\alpha = (1/2)$ in constrained i.i.d. mobility.

Proof: See Appendix. ■

Now, we can derive an upper bound on $E(\mathcal{N}(t))$.

Theorem 2: $E(\mathcal{N}(t)) \leq \alpha t$, where α is a constant determined by mobility model and $\alpha = 1/2$ under 2-D constrained i.i.d. mobility.

Proof: Denote by T the time interval that a disseminator transmits its active message to its next-hop disseminator. Clearly, random variable T stochastically dominates their first passage time T_F, which means $P(T > t) \geq P(T_F > t)$. Upon Lemma 3, there exists constant $C > 0$ such that

$$P(T > t) \geq Ce^{-\alpha t}, \text{ for all sufficiently large } t. \tag{15}$$

Therefore, $\mathcal{N}(t)$ is stochastically dominated by Poisson process with parameter α, and $E(\mathcal{N}(t)) \leq \alpha t$ accordingly, where $\alpha = 1/2$ under 2-D constrained i.i.d. mobility model. ■

As T_F corresponds to the residual life time of the *inter-meeting time* T_I, when T_I has a finite mean, T_F has the equilibrium distribution of T_I. Extensive existing studies have shown that T_I exhibits exponential tail decay under existing mobility models (such as random direction, random waypoint, and Brownian motion) in a bounded domain, while power-law decay in empirically traces as well as in infinite domain (See [30] and references inside). Because (15) holds under exponential as well as power-law decayed T_F with α determined by specific mobility model, Theorem 2 likely holds under other mobility models and in realistic mobility traces.

Equipped with previous results on movement vector $Y_M(t)$, propagation vector $Y_P(t)$, and $\mathcal{N}(t)$, we are ready to analyze dissemination distance for both *direction-invariant* dissemination and *geographic-assisted* dissemination strategies in Definition 3.

Lemma 4: In a 1-copy message dissemination (i.e., $L = 1$), $E(|D(t)|)$ is upper bounded by $\sqrt{tf_1(r, \lambda, a, \alpha)}$ for direction-invariant dissemination strategy, by $\sqrt{tf_2(r, \lambda, a, \alpha, t)}$ for geographic-assisted dissemination strategy, where $f_1(r, \lambda, a, \alpha)$ and $f_2(r, \lambda, a, \alpha, t)$ are shown in (16) and (17), respectively.

Proof: Denote $D(t) = (D_x(t), D_y(t)) = (\sum_{k=1}^{t} Y_x(k), \sum_{k=1}^{t} Y_y(k))$. Define by $Z(k)$ the event of active message jump at time k. Accordingly, $\mathcal{N}(t) = \sum_{k=1}^{t} 1_{Z(k)}$.

(i) Direction-invariant dissemination strategy means $E(Y_P^x(k)) = E(Y_P^y(k)) = 0$.

Because $E(Y_M^x(k)) = E(Y_M^y(k)) = 0$ under constrained i.i.d. mobility, and $Y_M(k)$ and $Y_P(k)$ are independent,

$$\begin{aligned} E\{D_x^2(t)\} &= E\left\{\sum_{k=1}^{t} [Y_M^x(k)]^2 + \left[Y_P^x(k) \cdot 1_{Z(k)}\right]^2\right\} \\ &= tE\left\{|Y_M^x(k)|^2\right\} + E\left(\mathcal{N}(t)\right) E\left\{|Y_P^x(k)|^2\right\}. \end{aligned}$$

Similar results can be obtained for $E\{D_y^2(t)\}$. Thus,

$$\begin{aligned} E\left(|D(t)|^2\right) &= E\left\{D_x^2(t) + D_y^2(t)\right\} \\ &= tE\left\{|Y_M(k)|^2\right\} + E\left(\mathcal{N}(t)\right) E\left\{|Y_P(k)|^2\right\}. \end{aligned}$$

Applying results in Lemmas 1 and 2, and Theorem 2, we get

$$E^2(|D(t)|) \leq E\left(|D(t)|^2\right) \leq t\left[a^2/2 + \alpha r^2 - \frac{\alpha}{\lambda\pi}\left(1 - e^{-\lambda\pi r^2}\right)\right].$$

Therefore, by denoting

$$f_1(r, \lambda, a, \alpha) = a^2/2 + \alpha r^2 - \frac{\alpha}{\lambda\pi}\left(1 - e^{-\lambda\pi r^2}\right), \tag{16}$$

we have $E(|D(t)|) \leq \sqrt{tf_1(r, \lambda, a, \alpha)}$.

(ii) Geographic-assisted dissemination strategy means $E(Y_P^x(k)) = E(Y_P^y(k)) \neq 0$. As $Y_M(k)$ and $Y_P(k)$ are independent and $E(Y_M^x(k)) = 0$,

$$\begin{aligned} E\left\{D_x^2(t)\right\} &= E\left\{\sum_{k=1}^{t}\left[\left(Y_M^x(k)\right)^2 + \left(Y_P^x(k) \cdot 1_{Z(k)}\right)^2\right]\right\} \\ &\quad + E\left\{\sum_{1 \leq k_1 < k_2 \leq t}\left[Y_P^x(k_1) \cdot 1_{Z(k_1)} \cdot Y_P^x(k_2) \cdot 1_{Z(k_2)}\right]\right\}. \end{aligned}$$

Similar result can be obtained for $E\{D_y^2(t)\}$. Hence,

$$\begin{aligned} E\left(|D(t)|^2\right) &\leq tE\left\{|Y_M(k)|^2\right\} \\ &\quad + \left(E(\mathcal{N}(t)) + \frac{t(t-1)}{2}\right) E\left\{|Y_P(k)|^2\right\}. \end{aligned}$$

In view of Lemmas 1 and 2, and Theorem 2,

$$E^2\left(|D(t)|\right) \leq E\left(|D(t)|^2\right) \leq tf_2(r, \lambda, a, \alpha, t),$$

where

$$f_2(r, \lambda, a, \alpha, t) = \frac{a^2}{2} + (\alpha + (t-1)/2)\left(r^2 - \frac{1 - e^{-\lambda\pi r^2}}{\lambda\pi}\right). \tag{17}$$

Therefore, $E(|D(t)|) \leq \sqrt{tf_2(r, \lambda, a, \alpha, t)}$. ■

Theorem 4 shows that the upper bounds of $E(|D(t)|)$ depend on node *velocity* (indicated by maximum movement length a per time slot), *mobility model* (represented by α), node *transmission ranger*, and node *density* λ. Furthermore, the expected dissemination distance can at most increase with the *square root* of time t in *direction-invariant* dissemination while approximately *linearly* with time t in *geographic-assisted* dissemination. The upper bound on dissemination distance under 1-copy geographic-assisted dissemination is approximately $\sqrt{t}$ times of that under 1-copy direction-invariant dissemination. In other words, comparing to direction-invariant dissemination, the increase in dissemination distance of utilizing geographic information accumulates as time goes by.

In L-copy message dissemination, number of disseminator is equal or less than L, i.e., $|\mathcal{V}(t)| \leq L$. Denote by $|D_i(t)|$ the distance between the source location and the location of the ith disseminator at time t and denote by dissemination distance $|D^L(t)|$ the maximum of $|D_i(t)|$.

Theorem 3: For L-copy direction-invariant dissemination, $E(|D^L(t)|) \leq (\sqrt{L-1} + 1)\sqrt{tf_1(r, \lambda, a, \alpha)}$; for L-copy geographic-assisted dissemination, $E(|D^L(t)|) \leq (\sqrt{L-1} + 1)\sqrt{tf_2(r, \lambda, a, \alpha, t)}$, where $f_1(r, \lambda, a, \alpha)$ and $f_2(r, \lambda, a, \alpha, t)$ are shown in (16) and (17), respectively.

Proof: To analyze $|D^L(t)|$, which equals the maximum of several random variables, we introduce Aven's [31] upper bound on the mean of the maximum of a number of random variables $\{Z_i, 1 \le i \le L\}$ with general distributions (not necessarily i.i.d.).

$$E(\max_{1\le i\le L} Z_i) \le \max_{1\le i\le L} E(Z_i) + \sqrt{\frac{L-1}{L}}\left(\sum_{i=1}^{L}\mathrm{Var}(Z_i)\right)^{1/2}. \tag{18}$$

Applying the above equation to $|D^L(t)|$

$$\begin{aligned} E\left(\left|D^L(t)\right|\right) &= E(\max_{v_i\in\mathcal{V}(t)}\{|D_i(t)|\}) \\ &\le \max_{1\le i\le L} E\left(|D_i(t)|\right) \\ &\quad + \sqrt{\frac{L-1}{L}}\left(\sum_{i=1}^{L} E\left(|D_i(t)|^2\right)\right)^{1/2} \end{aligned}$$

Based on the proof and results of Lemma 4, for L-copy direction-invariant dissemination,

$$E\left(\left|D^L(t)\right|\right) \le (\sqrt{L-1}+1)\sqrt{tf_1(r,\lambda,a,\alpha)} \tag{19}$$

for L-copy geographic-assisted dissemination,

$$E\left(\left|D^L(t)\right|\right) \le (\sqrt{L-1}+1)\sqrt{tf_2(r,\lambda,a,\alpha,t)}. \tag{20}$$

■

Remark 4: Regarding 1-copy dissemination as a base line, the upper bounds of dissemination distance increases $\sqrt{L-1}$ times under L-copy direction-invariant and geographic-assisted dissemination strategies, respectively.

B. Upper Bounds on Hitting Time

In 1-copy message dissemination, dissemination distance vector $D(t) = \sum_{k=1}^{t} Y(k)$, i.e., sum of i.i.d. random variables, we use martingale theory to study $\tau(d)$.

Lemma 5: Dissemination distance $\{|D(t)|^2\}_{t\in\mathbb{N}}$ is a submartingale with respect to Filtration $\mathcal{F}_t$, which is the σ-algebra generated by $\{D(k); k \le t\}$ for every $t \in \mathbb{N}$.

Proof: We prove that $|D(t)|^2$ is a submartingale by proving that $(D_x^2(t), D_y^2(t))$ is a 2D sub-martingale according to the following definition. A sub-martingale is defined as an integer-time stochastic process $\{Z_n; n \ge 1\}$ with the properties that $E[|Z_t|] < \infty$ for all $t \ge 1$ and

$$E[Z_t|Z_{t-1}, Z_{t-2}, \ldots, Z_1] \ge Z_{t-1};\ \text{for all } t \ge 2. \tag{21}$$

Denote dissemination vector $D(t) = (D_x(t), D_y(t)) = (\sum_{k=1}^{t} Y_x(k), \sum_{k=1}^{t} Y_y(k))$.

(i) Due to limited transmission range r and movement length a, $|Y_x(k)| \le r + a < \infty$. Then, for any $k \in \mathbb{N}$.

$$|D_x(t)| \le |Y_x(1)| + \ldots + |Y_x(t)| \le t \times (r+a) < \infty. \tag{22}$$

Similarly, $|D_y(t)| \le t \times (r+a) < \infty$.

(ii) Then we prove (21), which distinguishes martingale from other processes. Assume node v_i is the disseminator at time $t-1$. For any $t \in \mathbb{N}$, it holds that $E(D_x(t)|\mathcal{F}_{t-1}) = D_x(t-1) + E(Y_x(t))$ and $E(D_y(t)|\mathcal{F}_{t-1}) = D_y(t-1) + E(Y_y(t))$.

(a) When $Y(t) = Y_M(t)$, $Y_x(t) = Y_M^x(t) = A(t)\cos(\theta(t))$ and $Y_y(t) = Y_M^y(t) = A(t)\sin(\theta(t))$. In constrained i.i.d. mobility, $P(Y_M^x(t) = z) = P(Y_M^x(t) = -z)$ and $P(Y_M^y(t) = z) = P(Y_M^y(t) = -z)(0 \le z \le a)$. Hence, $E(Y_x(t)) = E(Y_M^x(t)) = 0$ and $E(Y_y(t)) = E(Y_M^y(t)) = 0$.

(b) When $Y(t) = Y_P(t) + Y_M(t)$, for direction-invariant dissemination, $E(Y_P^x(t)) = E(Y_P^y(t)) = 0$; for geographic-assisted dissemination, $E(Y_P^x(t)) \ge 0$ if $D_x(t) \ge 0$ and $E(Y_P^x(t)) \le 0$ if $D_x(t) \le 0$ (the same for $Y_P^y(t)$).

From (a) and (b), when $D_x(t) \ge 0$, $E(D_x(t)|\mathcal{F}_{t-1}) \ge D_x(t-1)$, which proves that $D_x(t)$ is submartingale. When $D_x(t) < 0$, $E(-D_x(t)|\mathcal{F}_{t-1}) \ge -D_x(t-1)$, which means that $-D_x(t)$ is submartingale. As $D_x^2(t) = D_x(t) * D_x(t) \cdot 1_{D_x(t)\ge 0} + (-D_x(t)) * (-D_x(t)) \cdot 1_{D_x(t)<0}$ and square function is convex, $D_x^2(t)$ is a submartingale. Similarly $D_y^2(t)$ is also a submartingale. Therefore, $|D(t)|^2 = D_x^2(t) + D_y^2(t)$ is a submartingale. ■

Upon Lemma 5, the hitting time $\tau^L(d)$ satisfies the following theorem.

Theorem 4: For L-copy direction-invariant dissemination, $P(\tau^L(d) \le t)$ is upper bounded by

$$\frac{t}{d^2}\left(f_1(r,\lambda,a,\alpha) + \sqrt{L-1}(r+a)\sqrt{f_1(r,\lambda,a,\alpha)}\right) \tag{23}$$

for L-copy geographic-assisted dissemination, $P(\tau^L(d) \le t)$ is upper bounded by

$$\frac{t}{d^2}\left(f_2(r,\lambda,a,\alpha,t) + \sqrt{L-1}(r+a)\sqrt{f_2(r,\lambda,a,\alpha,t)}\right). \tag{24}$$

Proof: According to L-copy message dissemination in Definition 2, the message is first spread to L distinct disseminators and then each of disseminators independently disseminates the message. Define $|D^{L^*}(t)|$ as the dissemination distance that L disseminators start to independently disseminate the message from time $t = 0$. As $|\mathcal{V}(t)| \le L$, $|D^L(t)|^2 = \max_{i\in\mathcal{V}(t)}\{|D_i(t)|^2\} \le |D^{L^*}(t)|^2 = \max_{i=1}^{L}\{|D_i(t)|^2\}$. Thus,

$$\begin{aligned} \tau^L(d) &= \inf_{t\ge 1}\{|D^L(t)| \ge d\} \\ &\ge \tau^{L^*}(d) = \inf_{t\ge 1}\{|D^{L^*}(t)| \ge d\}. \end{aligned} \tag{25}$$

Based on Lemma 5, $\{|D_i(t)|^2, 1 \le i \le L\}$ are independent sub-martingales. Hence, $|D^{L^*}(t)|^2 = \max_{i=1}^{L}\{|D_i(t)|^2\}$ is a sub-martingale. Based on Doob's Submartingale Maximal Inequality $P(\tau^{L^*}(d) \le t) \le (1/d^2)E(|D^{L^*}(t)|^2)$

$$\begin{aligned} P\left(\tau^{L^*}(d) \le t\right) &= P\left(\max_{1\le k\le t}\left|D^{L^*}(k)\right|^2 \ge d^2\right) \\ &\le \frac{1}{d^2}E\left(\left|D^{L^*}(t)\right|^2\right). \end{aligned} \tag{26}$$

Upon Aven's [31] upper bound on the mean of the maximum of a number of random variables in (18),

$$E\left(\left|D^{L^*}(t)\right|^2\right) = E\left(\max_{1\le i\le L}\left\{|D_i(t)|^2\right\}\right) \le \max_{1\le i\le L} E\left(|D_i(t)|^2\right) + \sqrt{\frac{L-1}{L}}\left(\sum_{i=1}^{L}\mathrm{Var}\left(|D_i(t)|^2\right)\right)^{1/2}.$$

Denote $Z = (|D_i(t)|^2/(r+a)^2 t)$. Clearly, $0 \le Z \le 1$. Thus

$$\mathrm{Var}\,(Z) = E(Z^2) - E^2(Z) \le E(Z)\,(1-E(Z)) \le E(Z)$$
$$\mathrm{Var}\left(|D_i(t)|^2\right) = (r+a)^4 t^2 \mathrm{Var}(Z) \le (r+a)^2 t E\left(|D_i(t)|^2\right). \tag{27}$$

Based on the proof of Lemma 4 and combining (25), (26), and (27), we complete our proof. ∎

Remark 5: By increasing L, L-copy dissemination can reduce dissemination latency as it can increase the probability of reaching nodes or infrastructures located d distance from the source within time t. Note that it is not known whether these upper bounds are achievable. The upper bounds may not be achievable by any algorithm in reality since the analysis is not based on realistic network and mobility models.

VI. Simulations and Applications

VANET is one of the promising applications of D2D communication. Many VANET applications require position-based vehicle-to-vehicle multicasting (e.g., disseminating traffic information to vehicles approaching an accident site). A natural match for this type of routing is geocasting protocols that forward messages to all nodes within an area of interest (AOI). Previous research work on geocast schemes for vehicular networks mostly propose various flooding-based schemes [28], [29]. One problem with a pure flooding-based geocasting protocol is that the flooding can cause network congestion [32]. *Limiting number of disseminators* and *geographic information exchanges* are two effective methods to reduce network load. However, there is a trade-off between minimizing network load and maximizing the spreading of the message by controling the number of disseminators and geographic information exchanges. It is not clear *how many disseminators are needed and whether geographic information exchanges should be used* for selecting relays.

Therefore, we perform simulations using the two general dissemination algorithms that we defined. Along with the simulation results and application requirements, we intend to provide guidelines on dissemination algorithm design for message dissemination in VANET. In this section, we first present and implement two dissemination algorithms in OMNet++ and compare simulation results with the analytic bounds presented in Sections IV and V. Then, we use two vehicle-to-vehicle communication scenarios to demonstrate the applications of our results.

A. Dissemination Algorithms

In *stateless opportunistic forwarding (SOF)* [17], a disseminator will choose next disseminator from its available neighbors at random. Stateless opportunistic forwarding has been suggested to be useful in intermittently connected networks [33]–[36]. It is particularly useful in vehicular *ad hoc* network as its global network topology is not known and rapidly varying due to high vehicle mobility and the presence or availability of the next-hop neighbors is not easily controllable. Clearly, SOF chooses next disseminator isotropically, thus is a type of *direction-invariant* dissemination. The SOF with one disseminator at each time slot is referred to as *1-copy SOF*. Similarly, dissemination algorithm that first sprays active messages to L disseminators and then each disseminator performs SOF independently, is referred to as *L-copy SOF*.

In *GPS-based broadcasting (GBB)* [18], a disseminator will choose its farthest neighbor as next disseminator so that a message can be spread out as fast as possible to certain locations (e.g., police station). GBB is useful for disseminating time-critical message (such as emergency warning) in VANETs. GBB is an example of *geographic-assisted* dissemination. The GBB with one disseminator is referred as *1-copy GBB*. Similarly, in *L-copy GBB*, source node first sprays active messages to L disseminators and then each disseminator performs GBB independently.

B. Simulation Results

1) Simulations Under Constrained Vehicle Mobility: Because currently there is no single benchmark of D2D communication network scenarios to evaluate its performance [37], we choose simulation parameters that are close to realistic network scenario and IEEE 1609/802.11p standards [38], [39] for VANETs. wIn a 10 km × 10 km network area, size of a small US city, 5000 nodes move according to constrained i.i.d. mobility model. Node density is $\lambda = 5 \times 10^{-5}$ vehicle/m^2. The average number of neighbors for each vehicle is equal to or less than $\pi * 200 * 200 * 5000/(10000 * 10000) = 2\pi$. As simulations in [22] suggested that six to eight neighbors can make a small size network connected with high probability, the above simulation settings will produce an intermittently connected network. Using the standard simulator 802.11p in OMNeT++INET framework, we set $carrierFrequency =$ 5.9 GHz, $wlan.opMode =$"p", $bitrate = 27$ Mbps, and $messageLength = 512$ B. The transmission range of node is $r = 200$ m. Each time slot is 1 second and the maximum movement length $a = 20$ m per time slot, which means that speed limit is about 45 mph. $L = 1$ or 4 for L-copy SOF and L-copy GBB.

As shown in Fig. 3, average dissemination distances of 1-copy SOF and 1-copy GBB over 15 simulation runs are upper bounded by bounds of expected dissemination distances of 1-copy direction-invariant and geographic-assisted dissemination in Lemma 4, respectively. Similarly, Fig. 4 shows that average dissemination distance of 4-copy SOF and 4-copy GBB are upper bounded by bounds of expected dissemination distances of L-copy direction-invariant and geographic-assisted

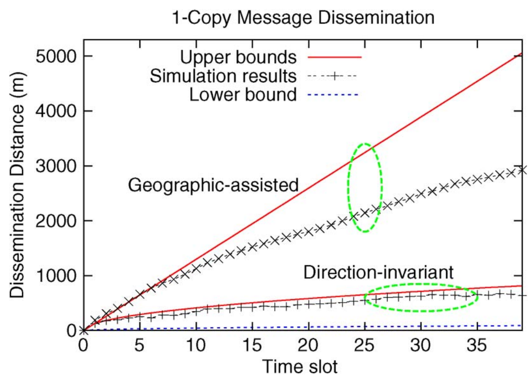

Fig. 3. Dissemination distance $|D(t)|$ of 1-copy direction-invariant and geographic-assisted message dissemination, respectively.

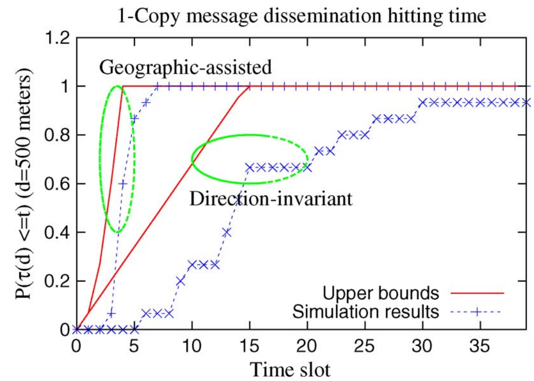

Fig. 5. $P(\tau(d) \leq t)(d = 500\ \text{m})$ in 1-copy direction-invariant and geographic-assisted message dissemination, respectively.

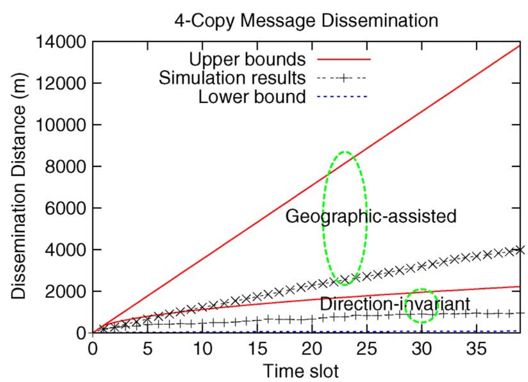

Fig. 4. Dissemination distance $|D(t)|$ of L-copy $(L = 4)$ direction-invariant and geographic-assisted message dissemination, respectively.

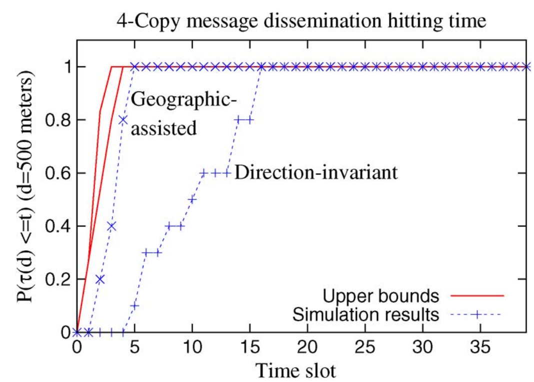

Fig. 6. $P(\tau(d) \leq t)(d = 500\,\text{m})$ in L-copy $(L = 4)$ direction-invariant and geographic-assisted message dissemination, respectively.

dissemination in Theorem 3, respectively. In a word, the average dissemination distances of SOF and GBB are well upper bounded by their corresponding analytic upper bounds.

As shown in Figs. 3 and 4, the upper bounds of expected dissemination distance under direction-invariant dissemination is tight, while there is a wide gap between the performance of GBB and the upper bounds under geographic-assisted dissemination. The gap could be lessened by more sophisticated algorithms, such as choosing nodes that move away from the source as relays.

More closely, Figs. 3 and 4 show that regardless of the number of disseminators, the expected dissemination distances of direction-invariant dissemination algorithms (i.e., 1-copy and 4-copy SOF) exhibit square root increase with time t, while that of geographic-assisted dissemination algorithms (i.e., 1-copy and 4-copy GBB) achieve approximately *linear* increase as time eclipses.

Both Figs. 3 and 4 reveal that comparing to direction-invariant dissemination, geographic-assisted dissemination significantly increases dissemination distance by utilizing geographic information. Increasing number of disseminators, although may benefit the dissemination reliability, is less effective than incorporating geographic-assisted dissemination in terms of enhancing dissemination distance.

Figs. 5 and 6 show that simulation results of $P(\tau(d) \leq t)$ of 1-copy and 4-copy message dissemination algorithms are well bounded by corresponding analytic bounds in Theorem 4. Both figures demonstrate benefits of utilizing geographic information in greatly reducing hitting time. However, geographic information becomes less effective in reducing hitting time when multiple disseminators are used (see Fig. 5) than when one disseminator is used (see Fig. 6). Increasing the number of disseminators seems to reduce hitting time in direction-invariant dissemination more dramatically than that in geographic assisted dissemination.

2) Simulations Under Highway Mobility: In a 20 km $\times$ 16 m rectangular network area, 200 nodes move according to highway mobility model. This setting characterizes a stretch of highway that has 4 lanes with each lane width 4 meters. As message dissemination on highway may affect several miles to dozens of miles, 20 km network length is suitable for studying message dissemination in highway scenario. Each time slot is 1 second. Each vehicle chooses a moving direction (left or right) at time 0 and will not change its moving direction at any time $t > 0$. The vehicle speed is uniformly distributed in [25 m/s, 35 m/s] (approximately 55 $\sim$ 80 mph). The transmission range $r = 250$ meters. The average number of neighbors for each vehicle is equal to or less than $2 * 250 * 200/20000 = 5$. Such simulation settings will produce an intermittently connected network with high probability [22].

Fig. 7 shows the average results of 1-copy SOF and GBB over 15 simulation runs as well as the theoretical bounds on dissemination distance of 1-copy message dissemination strategy in a highway scenario. Dissemination distances of SOF and GBB are upper bounded by the results on that of direction-invariant strategy and geographic-assisted strategy,

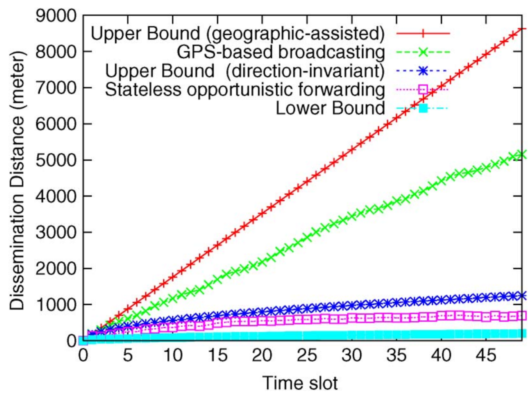

Fig. 7. Dissemination distance of 1-copy message dissemination strategies in a highway scenario.

respectively. Moreover, dissemination distance of SOF increases with square root of time t without using geographic information for disseminator selection. If Geographic information is used to speed up the dissemination, the dissemination distance increases linearly with time t. These results are consistent with our theoretical analysis as well as results in the previous subsection.

C. Applications

We choose two important applications in VANETs defined by IEEE 1609 Standard [38], which are *post-crash warning* and *emergency vehicle signal preemption* to show how our results could be used in guiding the network design. The application requirements are obtained from vehicle safety communication project report [40] by National Highway Traffic Safety Administration in Department of Transportation of U.S. In the following, we assume that the time is slotted and each time interval is 1 second as in other studies (e.g., paper [41]). Node speed, density, and transmission range are assumed to be the same with simulation settings of scenario 1) in the previous subsection.

1) Post-Crash Warning: In the application of post-crash warning, a disabled vehicle (due to an accident or mechanical breakdown) will warn approaching vehicles of this emergency and its position, and will discontinue broadcast when the accident is cleared. The allowable latency for this application is 5 sec [40].

Suppose vehicle speed is about 20 m/s and drivers need about 1.5sec to react and 3sec to brake if it is necessary, which mean that geocast needs to target vehicles within range of about 100 m. Simulation results (Figs. 3 and 4) show that in 5 time slots, dissemination distances are about 252 m, 396 m, 660 m, and 669 m for 1-copy and 4-copy SOF, 1-copy and 4-copy GBB, respectively. Since more sophisticated algorithms could achieve better performance than SOF and GBB algorithms, both direction-invariant and geographic-assisted algorithms could disseminate the message fast enough to reach the borders of targeted area.

As propagation speed is fast enough, dissemination strategy should focus on achieving the reliability requirement of this safety application. As simultaneous rebroadcasting of multiple disseminators can enhance the probability of vehicles receiving this warning, L-copy message dissemination strategy with $L > 1$ could be a good candidate in fast and reliable post-crash warning dissemination.

2) Emergency Vehicle Signal Preemption: Emergency vehicle signal preemption allows the emergency vehicles to override traffic signals. When an emergency vehicle is approaching an intersection, it initiates a message dissemination targeting vehicles around that intersection. After receiving the message and verifying that the request has been made by an authorized source, vehicles around the intersection should prepare to stop and provide the right of way to the emergency vehicle.

As an example, we give specific and reasonable data to illustrate a scenario of this application. We assume that the dissemination targets vehicles in the circular region around the intersection with radius 100 m. Suppose the emergency vehicle moves at speed about 20 m/s. In order to inform targeted vehicles about 15 sec before the emergency vehicle enters that region around the intersection, the emergency vehicle should initiate the dissemination when it is about 300 m away from that region. The allowable latency for this application is approximately 10 sec according to report [40]. Hence, the message should at least hit the farthest locations of targeted region, which is 500 m from the source, within 10 time slots.

From Figs. 5 and 6, the upper bounds of the probability of reaching 500 meters in 10 time slots are about 70% for 1-copy direction-invariant dissemination while 100% for other three dissemination strategies. Furthermore, we can see that the probability of reaching 500 meters in 10 time slots is about 30% and 40%, 100% and 100% for 1-copy and 4-copy SOF, 1-copy and 4-copy GBB, respectively. This means that direction-invariant dissemination is incapable of serving this application, while geographic-assisted dissemination could satisfy this application's requirements.

Remark 6: Dissemination strategies that use multiple disseminators are suitable for applications like post-crash warning, in which dissemination targets an area near the source location and requires high dissemination reliability. Dissemination strategies that utilize geographic information to choose relays are needed for applications like emergency vehicle signal preemption, which targets an area far from the source location.

VII. Conclusion

In this paper, we have studied the spatial and temporal limits of message dissemination in dynamic and intermittently connected D2D communication networks. By focusing on movement of active messages rather than specifying relays on information propagation paths, we have derived lower and upper bounds for the dissemination distance and hitting time. Simulation results of two dissemination algorithms validated our analysis. Two applications of vehicle-to-vehicle communication are presented to show that dissemination algorithms with multiple disseminators are suitable for message dissemination targeting area close to source location while dissemination algorithms assisted by geographic information are suitable for message dissemination targeting area far from the source.

APPENDIX

Proof of Poisson Node Distribution Under Constrained i.i.d. Mobility Model

Proof: Let the circular region $\mathcal{A}$ centered at a random selected location in the network with radius a be partitioned to $m+1$ concentric rings $\mathcal{A}_i$ indexed by $i, 0 \le i \le m$, each of which is with equal ring width ϵ. Note that $\mathcal{A}_0$ is a circle centered at the center of $\mathcal{A}$ with radius ϵ. Denote the area enclosed by ring $\mathcal{A}_i$ as $S_{\mathcal{A}_i}$, and define by $N_i(t)$ the number of nodes in $\mathcal{A}_i$ at time t. Since nodes are Poisson distributed in the network area initially, i.e., $N_i(0) \sim Pois(\lambda S_{\mathcal{A}_i})$.

Under constrained mobility model in Definition 1, when $t=1$, number of nodes in interval $\mathcal{A}_0$ is

$$N_0(1) = N_0(0) \int_{\mathcal{A}_0} \frac{1}{\pi a^2} d\mathcal{A}_0 + \ldots + N_m(0) \int_{\mathcal{A}_m} \frac{S_{\mathcal{A}_0}}{S_{\mathcal{A}_m}} \frac{1}{\pi a^2} d\mathcal{A}_m. \tag{28}$$

Denote $p_i \triangleq \int_{\mathcal{A}_i} (S_{\mathcal{A}_0}/S_{\mathcal{A}_i})(1/\pi a^2) d\mathcal{A}_i, i = 0, 1, \ldots, m$, and $N_i^*(0) \triangleq p_i N_i(0)$. Accordingly

$$\begin{aligned} &P\left(N_i^*(0) = k^*\right) \\ &= \sum_{k \ge k^*} \binom{k}{k^*} p_i^{k^*} (1-p_i)^{k-k^*} \cdot e^{-\lambda S_{\mathcal{A}_i}} \frac{(\lambda S_{\mathcal{A}_i})^k}{k!} \\ &= e^{-\lambda S_{\mathcal{A}_i}} \frac{(p_i \lambda S_{\mathcal{A}_i})^{k^*}}{k^*!} \sum_{k \ge k^*} \frac{[(1-p_i)\lambda S_{\mathcal{A}_i}]^{k-k^*}}{(k-k^*)!} \\ &= e^{-p_i \lambda S_{\mathcal{A}_i}} \frac{(p_i \lambda S_{\mathcal{A}_i})^{k^*}}{k^*!}. \end{aligned} \tag{29}$$

$N_i^*(0)$ follows Poisson distribution with parameter $p_i \lambda S_{\mathcal{A}_i}$. Since $\{N_i^*(0), i = 0, \ldots, m\}$ are independent Poisson random variables and $N_0(1)$ equals to the sum of $N_i^*(0)$,

$$N_0(1) \sim Pois\left(\sum_{i=0}^{m} p_i \lambda S_{\mathcal{A}_i}\right) \sim Pois(\lambda S_{\mathcal{A}_0}). \tag{30}$$

Similarly, the number of nodes in any small circular region in the network at $t=1$ follows the same Poisson distribution as at $t=0$. Assume $N_i(t) \sim Pois(\lambda S_{\mathcal{A}_i}), 0 \le i \le m$, using the same method as above, we can show that $N_i(t+1) \sim Pois(\lambda S_{\mathcal{A}_i}), 0 \le i \le m$. Therefore, by induction, nodes are Poisson distributed in the network area at all times. ■

Proof of the Lemma 3:

Proof: We prove this lemma using the same methodology that is used to derive inter-meeting time of two nodes under 2-D isotropic random walk in [30]. According to Definition 1, the position of a node v at time t can be written as

$$X_v(t) - X_v(0) = \sum_{k=1}^{t} Y_M^v(k) = \sum_{k=1}^{t} A_v(k) e^{i\theta_v(k)} \tag{31}$$

where $Y_M^v(k)$ is the movement vector at time k. Define by $C(t) = X_u(t) - X_v(t)$ the difference vector between the positions of nodes u and v at time t. Under 2-D constrained i.i.d. node mobility, we observe that

$$C(t) = \sum_{k=1}^{t} \left(A_u(k) e^{i\theta_u(k)} - A_v(k) e^{i\theta_v(k)}\right). \tag{32}$$

Under constrained i.i.d. mobility, $A_u(k)$ and $A_v(k)$ are all i.i.d. and so are $\theta_v(k)$ and $\theta_v(k)$. Thus, $A_u \cos(\theta_u) - A_v \cos(\theta_v)$ is symmetric and continuous (because uniform distribution is continuous). Accordingly, $[C(t)]_x$, sum of random variables $A_u(k) \cos\theta_u(k) - A_v(k) \cos\theta_v(k)$ for $1 \le k \le t$, is 1-D random walk. According to results in [30], $P(T_F > t) \sim t^{-1/2}$, thus completes our proof. ■

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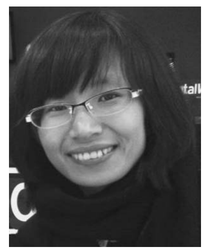

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